

Modeling and Computation of Electric and Magnetic Fields

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Maxwell's Equations in free space - overview

Differential form / *local form*

Divergence equations

$$\nabla \cdot \mathbf{E} = \rho / \epsilon_0 \qquad \nabla \cdot \mathbf{B} = 0$$

Curl equations

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \qquad \nabla \times \left(\frac{\mathbf{B}}{\mu_0} \right) = \mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

Two new quantities:

ρ : volume charge density [C/m⁺³]

$\mathbf{J}$: current density [A/m⁺²]

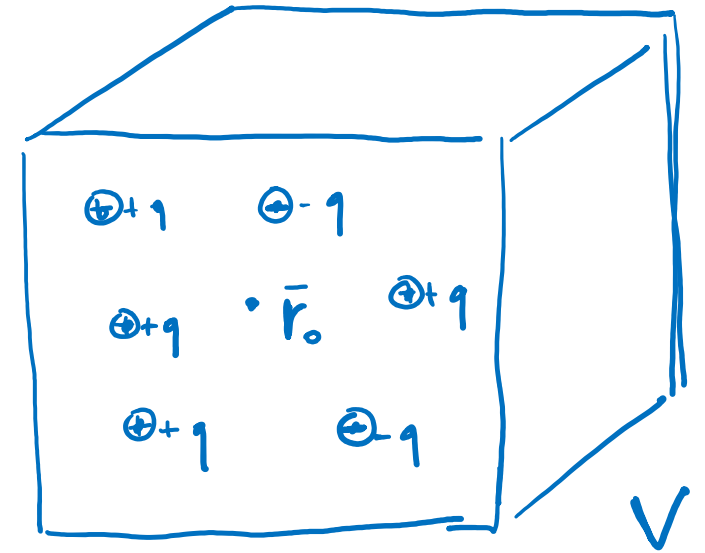
Volume charge density

Let $\mathbf{r}_0$ be a point in a region V with volume ΔV

$$\Delta Q = \sum_i q_i \in V$$

$\rightarrow \rho(\mathbf{r}_0) = \lim_{\Delta V \rightarrow 0} \frac{\Delta Q}{\Delta V} \quad [\text{C/m}^3]$

ΔQ : net electric charge within V



Note: **macroscopic** limit.

ΔV *small enough* to provide point information in the macroscopic scale

ΔV *large enough* to provide produce average of electric charge around $\mathbf{r}_0$

Current density

Let $\mathbf{r}_0$ be a point on a surface S with area ΔS

component of $\vec{J}$
along $\hat{n}$

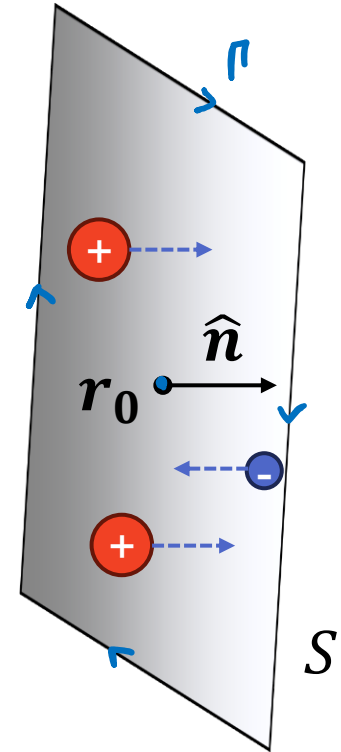
$$\mathbf{J} \cdot \hat{n}(\mathbf{r}_0) = \lim_{\substack{\Delta S \rightarrow 0 \\ \Delta t \rightarrow 0}} \frac{\Delta Q}{\Delta S \Delta t}$$
$$\Delta Q = \sum_i q_i \in V$$
$$\left[\frac{\text{C}}{\text{m}^2 \text{s}} = \frac{\text{A}}{\text{m}^2} \right]$$

ΔQ : net electric charge through S during Δt

Note: *macroscopic* limits.

$\Delta S, \Delta t$ *small enough* to provide point information in the macroscopic scale

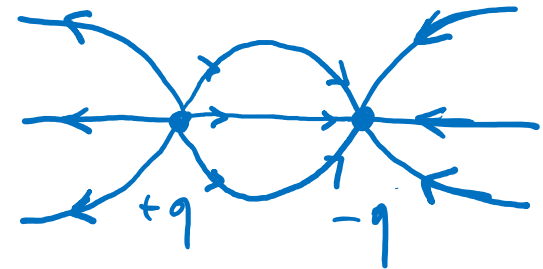
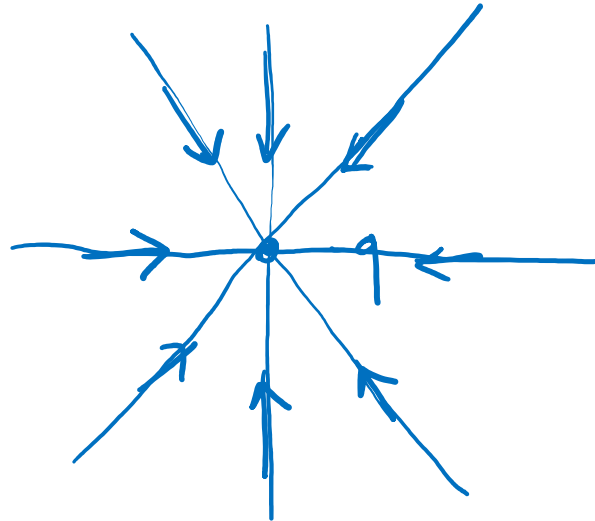
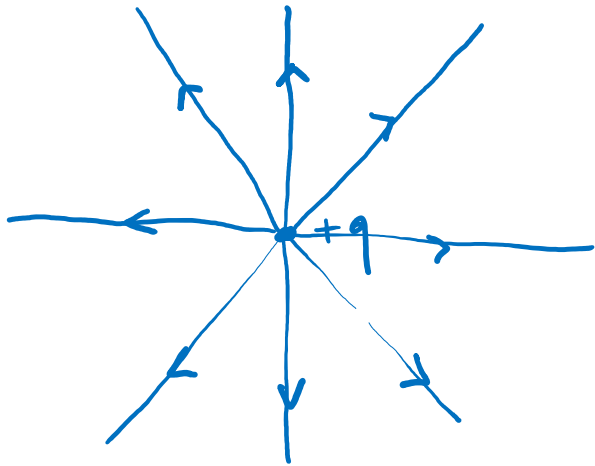
$\Delta S, \Delta t$ large enough to provide produce averages around r_0



Gauss law for E-field – local form

$$\rightarrow \boxed{\nabla \cdot \mathbf{E} = \rho / \epsilon_0}$$

Electric field lines originate on positive charges and terminate on negative charges



Gauss law for E-field – integral form (derivation)

$$\nabla \cdot \mathbf{E} = \rho / \epsilon_0$$

integrate both sides over volume V

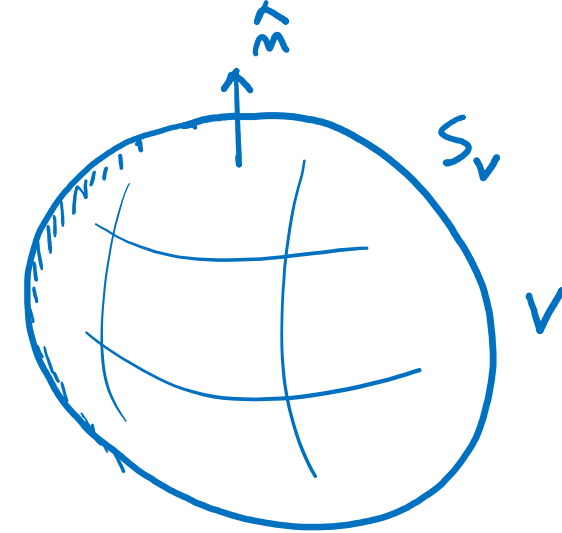
$$\int_V \nabla \cdot \mathbf{E} dV = \int_V \frac{\rho}{\epsilon_0} dV$$

apply divergence theorem

$$\oint_{S_V} \mathbf{E} \cdot \hat{\mathbf{n}} dS = \frac{1}{\epsilon_0} \int_V \rho dV$$

$$Q_{\text{enc}} = \int_V \rho dV$$

$$\boxed{\oint_{S_V} \mathbf{E} \cdot \hat{\mathbf{n}} dS = \frac{Q_{\text{enc}}}{\epsilon_0}}$$

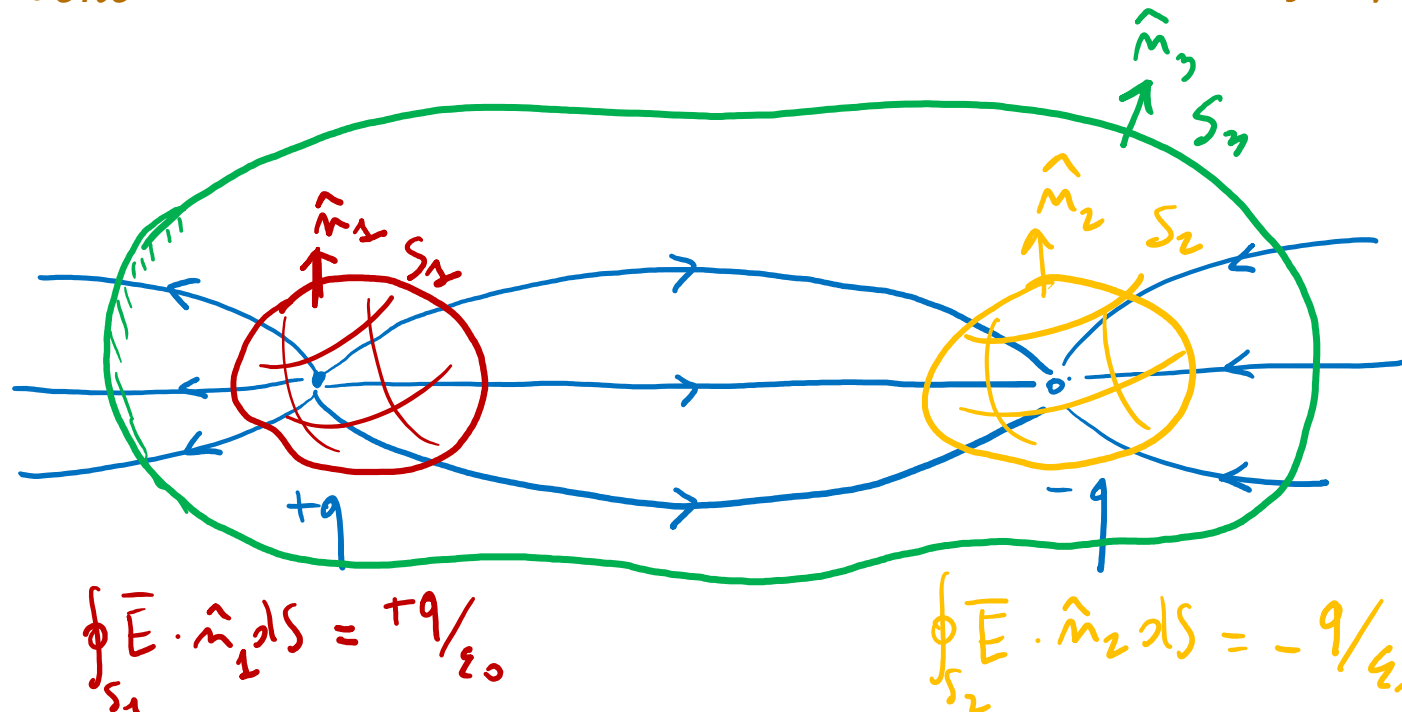


Gauss law for E-field – integral form (interpretation)

Gauss law for E – integral form

$$\oint_{S_V} \mathbf{E} \cdot d\mathbf{S} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

The flux of $\mathbf{E}$ through any closed surface S_V is proportional to the net charge Q_{enc} enclosed within the volume V bounded by S_V

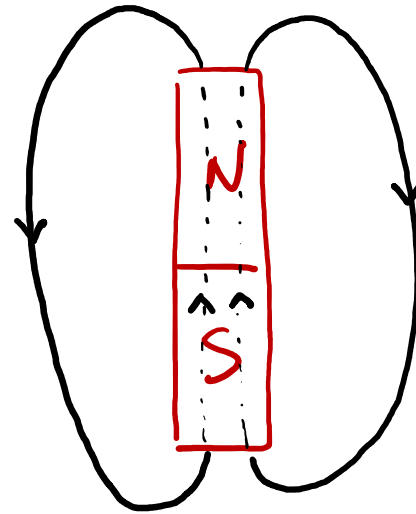


Gauss law for B-field – local form

→ $\nabla \cdot \mathbf{B} = 0$

Magnetic field lines have no sources or sinks; they form closed loops

Example: field lines of a permanent magnet



Gauss law for B-field – integral form (derivation)

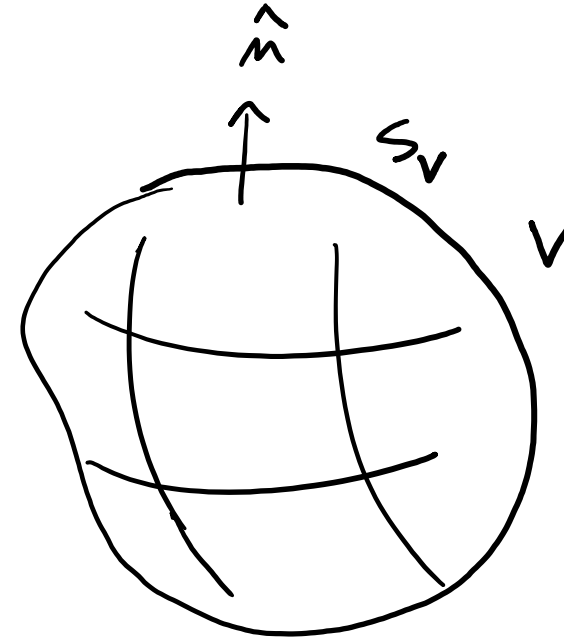
$$\rightarrow \nabla \cdot \mathbf{B} = 0$$

integrate both sides over volume V

$$\int_V \nabla \cdot \mathbf{B} dV = 0$$

apply divergence theorem

$$\oint_{S_V} \mathbf{B} \cdot \hat{\mathbf{n}} dS = 0$$



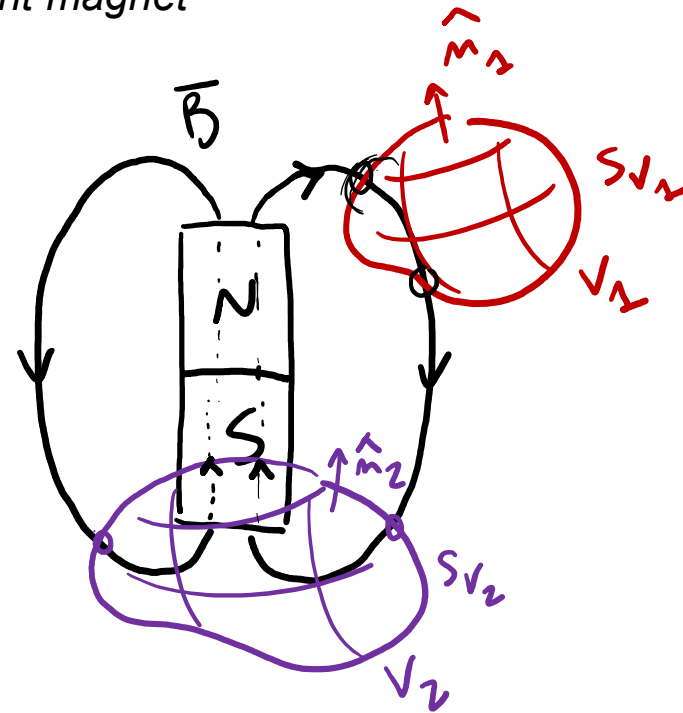
Gauss law for B-field – integral form (interpretation)

Gauss law for B – integral form

$$\oint_{S_V} \mathbf{B} \cdot \hat{\mathbf{n}} dS = 0$$

*The flux of $\mathbf{B}$ through any closed surface S_V is null $\rightarrow \mathbf{B}$ is **solenoidal***

Example: flux evaluation for permanent magnet



$$\oint_{S_{V_1}} \mathbf{B} \cdot \hat{\mathbf{n}}_1 = 0$$

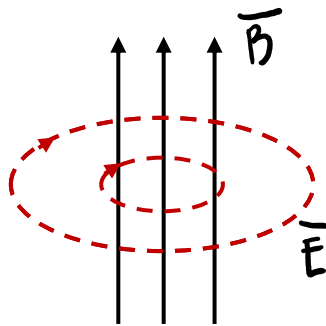
$$\oint_{S_{V_2}} \mathbf{B} \cdot \hat{\mathbf{n}}_2 = 0$$

Faraday-Neumann-Lenz law – local form

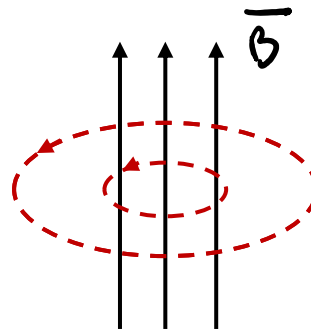
$$\rightarrow \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

A time-varying magnetic field produces a circulating electric field

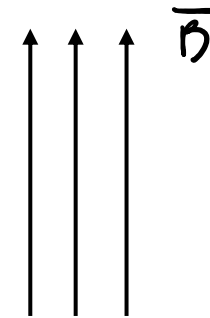
$$\frac{\partial \mathbf{B}}{\partial t} > 0$$



$$\frac{\partial \mathbf{B}}{\partial t} < 0$$



$$\frac{\partial \mathbf{B}}{\partial t} = 0$$



Faraday-Neumann-Lenz law – integral form

Consider open surface S bounded by curve Γ_S

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

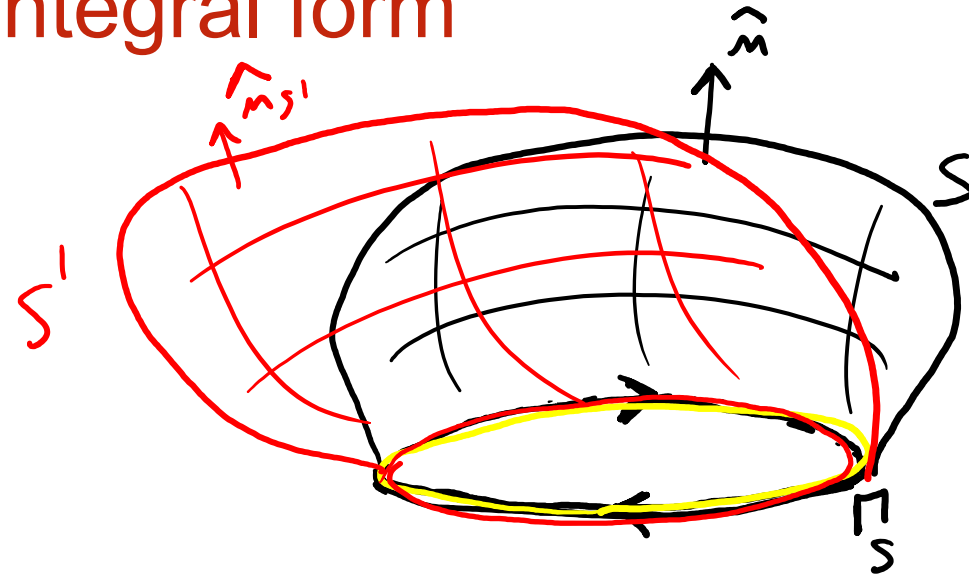
integrate both sides over S

$$\int_S \nabla \times \mathbf{E} \cdot \hat{\mathbf{n}} dS = \int_S -\frac{\partial \mathbf{B}}{\partial t} \cdot \hat{\mathbf{n}} dS$$

apply Stokes theorem

$$\oint_{\Gamma_S} \mathbf{E} \cdot \hat{\mathbf{l}} dS = -\frac{d}{dt} \int_S \mathbf{B} \cdot \hat{\mathbf{n}} dS$$

$$\boxed{\varepsilon = -\frac{d}{dt} \Phi_S(\mathbf{B})} = -\frac{d}{dt} \Phi_{\Gamma_S}(\bar{\mathbf{B}})$$



[LINKED FLUX]

Flux of ANY solenoidal
VECTOR FIELD depends
only on boundary of S

$$\Phi_S(\bar{\mathbf{B}}) = \Phi_{S'}(\bar{\mathbf{B}})$$

ε [V]: **electromotive force**

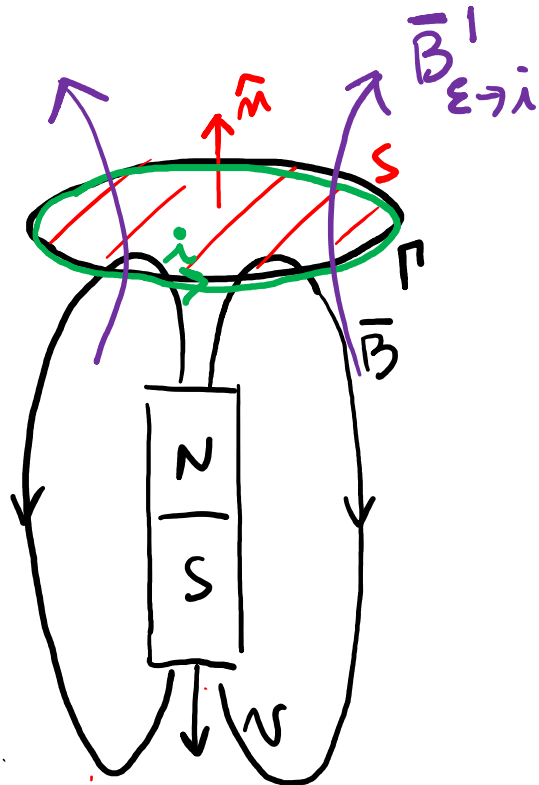
Work per unit charge done by a non-conservative electric field around a closed path

Faraday-Neumann-Lenz law – integral form

$$\varepsilon = - \frac{d}{dt} \Phi_s(\vec{B})$$

A time-varying magnetic flux through a surface produces a circulation of the electric field around its boundary

Example: FNL for moving magnet



$$v = 0 \quad \underbrace{\Phi_s(\vec{B})}_{\int_S \vec{B} \cdot \hat{n} dS} \neq 0 \Rightarrow \frac{d}{dt} \Phi_s(\vec{B}) = 0$$

$$v \neq 0 \quad \Phi_s(\vec{B}) \neq 0 \Rightarrow \boxed{\frac{d}{dt} \Phi_s(\vec{B}) < 0}$$

↓ ON WIRE

$$\oint_{\Gamma} \vec{E} \cdot \hat{e} d\ell \neq 0 \Rightarrow \varepsilon$$

↓ $\vec{B}'$

$$\boxed{\Phi_s(\vec{B}') > 0}$$

A note on electric field

Electric field produced by charges

$$\nabla \times \mathbf{E} = 0$$

Conservative field

Can be written as $\mathbf{E} = -\nabla V$

Zero circulation around any closed path

Electric field produced by electromagnetic induction

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Non-conservative field

Non-zero circulation may exist around closed paths

Charges produce the **conservative part** of the electric field, while time-varying magnetic fields produce a **non-conservative circulating component**

Ampere-Maxwell law – local form

$$\nabla \times \left(\frac{\mathbf{B}}{\mu_0} \right) = \mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

The total current density produces a circulating magnetic field

Ampere-Maxwell law – integral form

Consider open surface S bounded by curve Γ_S

$$\nabla \times \left(\frac{\mathbf{B}}{\mu_0} \right) = \mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

integrate both sides over S

apply Stokes theorem

$$\boxed{\mathcal{F} = i_T}$$

$\mathcal{F}$ [A]: **magnetomotive force**

circulation of magnetic field intensity
along a closed path

Faraday-Neumann-Lenz law – integral form

$$\boxed{\mathcal{F} = i_T}$$

An electric current through a surface produces a circulation of the magnetic field intensity around its boundary

Exercise: show that $\mathbf{J}_T$ is a solenoidal vector field

Current continuity equation